GEN-101 — General Course Syllabus

OFFICIAL COURSE SYLLABUS & CURRICULUM GUIDE

Course Code: GEN-101	**Category:** Computer Science & Tech
Academic Term: 2026 Academic Year	**Format:** E-Learning / Self-Paced
Prerequisites: Basic Problem Solving	**Estimated Hours:** 40 Hours Content

1. Course Description & Learning Objectives

Welcome to **General Course Syllabus (GEN-101)**! This course is designed to equip students with deep theoretical understanding and practical hands-on proficiency. By completing this curriculum, learners will master key concepts, build industry-standard projects, and develop problem-solving skills necessary for professional software development.

Key Learning Outcomes:

- Master foundational and advanced topics through structured video lectures and interactive materials.
- Apply theoretical principles to real-world coding projects and algorithmic problem solving.
- Understand best practices, debugging strategies, and architectural design patterns.
- Pass comprehensive evaluations and earn official course certification.

2. Course Modules & Detailed Schedule

Module / Topic	Key Topics & Assignments	Weightage
Module 1: Foundations & Environment Setup	Introduction to key concepts, installing dependencies, basic syntax and initial projects.	25%
Module 2: Core Data Structures & Control Flow	Conditionals, loops, functions, lists, dicts, memory management and standard libraries.	30%
Module 3: Object-Oriented & Modular Architecture	Classes, inheritance, encapsulation, design patterns, and package management.	35%
Module 4: Practical Applications & Capstone Project	Building end-to-end applications, testing, optimization, and final deployment.	40%

3. Grading Policy & Assessment Breakdown

Component	Percentage	Details
Practical Exercises & Quizzes	30%	Weekly online quizzes & code exercises
Mid-Term Assessment	30%	Comprehensive theoretical & coding test
Final Capstone Project	40%	Individual end-to-end practical project